Impact of the COVID-19 pandemic on the performance of endoscopic examinations and gastrointestinal cancer mortality in Maranhão

INTRODUCTION

Cancer is one of the main causes of morbidity and mortality in the world, especially among individuals under 70 years old, regardless of the Human Development Index of countries, being recognized as a relevant public health problem. In addition, the prevalence profile of different types of cancer has been changing over time, possibly related to changes in socioeconomic conditions, the urbanization process and the adoption of unhealthy lifestyle habits, such as sedentary behavior, inadequate diet and chemical dependency[1].

Given this, cancer represents the second main cause of death in the world, surpassed only by cardiovascular diseases. Among the different types, gastrointestinal tract cancers present high mortality, with estimates of more than 153 thousand deaths in the United States in 2016 and more than 169 thousand in 2021. In Brazil, this group of neoplasms also demonstrates an important impact on public health, being responsible for more than 81 thousand deaths in 2021[2].

In this context, upper gastrointestinal endoscopy is currently one of the main examinations used in the evaluation of diseases of the gastrointestinal tract, playing a fundamental role both in the diagnosis and treatment of alterations of the esophagus, stomach and duodenum. In addition, early performance of the examination allows the identification of diseases in initial stages, such as cancer, inflammatory diseases and peptic ulcer, favoring more effective management and better prognosis[3].

During the COVID-19 pandemic, there were profound repercussions on health systems worldwide, promoting overload of hospital services, reorganization of health care and a decrease in the performance of diagnostic and therapeutic procedures classified as non emergency. In Brazil, this scenario made it more difficult for the population to access specialized services, including endoscopic examinations and oncology related care, which may have contributed to the diagnosis of cancer cases at more advanced stages and to a potential increase in mortality[4].

Studies conducted during the pandemic showed that many oncology patients delayed seeking specialized care, mainly due to fear of COVID-19 contamination in health services. In Maranhão, a qualitative study based on patient perceptions in an oncology outpatient clinic

reported negative impacts on medical follow up and access to oncology services during the pandemic, which may be relevant to delays in the early diagnosis of gastrointestinal cancers specifically, a condition that can compromise prognosis and reduce the chances of effective treatment[5].

Unfortunately, studies evaluating the relationship between the reduction in endoscopic examinations during the COVID-19 pandemic and gastrointestinal cancer mortality in Maranhão are still scarce. Therefore, this study seeks to analyze the impact of the COVID-19 pandemic on the performance of endoscopic examinations and gastrointestinal cancer mortality in the state of Maranhão.

METHODS

This study is a retrospective, ecological investigation based on secondary, publicly available, aggregated data. Monthly data on ambulatory endoscopic production and gastrointestinal cancer mortality in the state of Maranhão, Brazil, were obtained from the Outpatient Information System (SIA/SUS) and the Mortality Information System (SIM), both maintained by the Department of Informatics of the Brazilian Unified Health System (DATASUS), accessed through the TabNet query platform. The analytic window comprised January 2018 to December 2024, totaling 84 months.

Ambulatory production of upper gastrointestinal endoscopy (EGD, procedure code 0209010037) and colonoscopy (procedure code 0209010029) was extracted by month. Deaths from malignant neoplasms of the gastrointestinal tract were extracted by month from SIM, using the corresponding International Classification of Diseases, 10th revision categories for esophageal, gastric and colorectal cancer. All counts represent total approved production or registered deaths for the state of Maranhão, without population adjustment, consistent with the time-series design of the analysis.

The onset of the COVID-19 pandemic was defined as March 2020, anchored to Decreto Estadual nº 35.672/2020, issued on March 19, 2020, which declared a state of public health emergency in Maranhão. This externally validated, legally defined date was adopted as the primary breakpoint

to avoid a data-driven selection of the intervention point, and April 2020 was used as an alternative breakpoint in a sensitivity analysis.

The effect of the pandemic on EGD and colonoscopy production was estimated using segmented interrupted time series regression, with terms for the pre-pandemic trend, the immediate level change at the breakpoint, and the change in the post-pandemic slope. Because ordinary least squares residuals showed first-order autocorrelation, models were fitted by generalized least squares with a first-order autoregressive error structure, implemented in R using the gls function of the nlme package. Gastrointestinal cancer mortality was modeled using an analogous segmented specification, with the same pre-trend, level-change and slope-change terms, fitted by negative binomial regression using the glm.nb function of the MASS package to account for overdispersion in the monthly death counts, and results are reported as incidence rate ratios. The temporal association between endoscopic production and mortality was explored using the cross-correlation function applied to the first-differenced series, to assess whether changes in procedure volume preceded or lagged changes in mortality across multiple monthly lags. A two-sided $p<0.05$ was considered statistically significant, and residual diagnostics included the Ljung-Box test for residual autocorrelation.

This study used only secondary, aggregated, publicly available data without individual patient identification, and was therefore exempt from evaluation by a research ethics committee, in accordance with Resolução CNS nº 510/2016, Article 1. No external funding was received.

RESULTS

Four main findings emerged from this analysis. Endoscopic production in Maranhão fell abruptly at the onset of the pandemic, with reductions exceeding 80% for both procedures in the second quarter of 2020. Production recovered above pre-pandemic levels within two years. Gastrointestinal cancer mortality showed no corresponding disruption. No lagged association was detected between endoscopic volume and mortality.

Production

Ambulatory EGD production grew from 28,984 examinations in 2000 to a pre-pandemic peak of 43,665 in 2007. It then fluctuated between 24,740 and 43,291 examinations per year through 2019. Colonoscopy volume, recorded from 2008 onward, rose from 1,971 to 4,448 examinations over the same interval.

The second quarter of 2020 marked the point of maximum disruption. EGD production fell 80.7% (8,759 to 1,687 procedures) relative to the same quarter of 2019. Colonoscopy fell 81.7% (1,032 to 189 procedures). Both procedures exceeded their 2019 totals by 2024, EGD reaching 44,157 examinations and colonoscopy reaching 7,009.

The segmented interrupted time series model, using the March 2020 cutoff, estimated a level change of -556.8 procedures for EGD (p=0.162) and -96.4 for colonoscopy (p=0.148). Post-pandemic slope changes were +46.57 procedures per month for EGD (p=0.084) and +4.11 for colonoscopy (p=0.313) (Table 1). Using the April 2020 sensitivity cutoff, level changes became statistically significant for both procedures (EGD: -1549.7, p=0.0003; colonoscopy: -164.8, p=0.016).

Table 1. Segmented interrupted time series analysis of EGD and colonoscopy production, Maranhão, 2018–2024 (primary cutoff: March 2020).

Parameter	EGD, β (p)	Colonoscopy, β (p)
Pre-pandemic trend (β1, per month)	-12.68 (p=0.597)	+2.09 (p=0.580)
Immediate level change (β2)	-556.8 (p=0.162)	-96.4 (p=0.148)
Post-pandemic slope change (β3, per month)	+46.57 (p=0.084)	+4.11 (p=0.313)
Immediate level change, April 2020 sensitivity cutoff (β2)	-1549.7 (p=0.0003)	-164.8 (p=0.016)

AR(1) autocorrelation coefficient: φ=0.611 (EGD), φ=0.472 (colonoscopy). Model estimated by generalized least squares with first-order autoregressive errors. Primary cutoff anchored to Decreto Estadual nº 35.672/2020 (March 19, 2020); April 2020 shown as sensitivity analysis.

Mortality

Gastrointestinal cancer deaths increased 26.5% between 2018 and 2024 (614 to 777 deaths), an average annual growth rate of 4.0%. The segmented negative binomial model estimated a

level-change incidence rate ratio of 0.780 at the March 2020 cutoff (p<0.001). The post-pandemic slope change was not significant (p=0.950) (Table 2).

Table 2. Segmented negative binomial regression of gastrointestinal cancer mortality, Maranhão, 2018–2024 (March 2020 cutoff).

Parameter	β (p)	IRR
Pre-pandemic trend (β1, per month)	+0.0069 (p=0.055)	1.007
Immediate level change (β2)	-0.2487 (p<0.001)	0.780
Post-pandemic slope change (β3, per month)	-0.0002 (p=0.950)	0.9998

IRR: incidence rate ratio (exp β). Model estimated by maximum likelihood, dispersion parameter estimated jointly with regression coefficients.

Association

The cross-correlation function between the differenced endoscopy and mortality series showed no coherent pattern across the lags tested. Four of 49 lag-correlation pairs exceeded the 95% significance bounds, consistent with the number expected by chance under multiple comparisons.

DISCUSSION

This study found no statistically significant immediate change in the volume of upper gastrointestinal endoscopy (EGD) or colonoscopy in Maranhão at the onset of the COVID-19 pandemic, using March 2020 (Decree No. 35,672/2020) as the pre-specified breakpoint in a segmented interrupted time series with AR(1) errors. The estimated level change was -556.8 procedures for EGD (p=0.162) and -96.4 procedures for colonoscopy (p=0.148), and the post-pandemic slope changes were likewise not significant (EGD: +46.57 per month, p=0.084; colonoscopy: +4.11 per month, p=0.313). A sensitivity analysis using April 2020 as an alternative, data-driven breakpoint yielded markedly larger and statistically significant level drops (EGD: -1549.7, p=0.0003; colonoscopy: -164.8, p=0.0163), indicating that the magnitude and significance of the estimated disruption are sensitive to the exact date chosen as the onset of the interruption, and should be interpreted with this limitation in mind rather than as a single definitive estimate.

These findings are broadly consistent with the international literature describing a sharp, transient reduction in endoscopic and cancer screening activity during the early pandemic period, although the magnitude reported elsewhere has generally been larger than what was estimated here for Maranhão. A systematic review by Mazidimoradi et al.[6] reported that colorectal cancer screening activity decreased between 28% and 100% across different health systems during the initial pandemic months, with completion colonoscopies falling by approximately 65.7% and surveillance colonoscopies by 44.6% to 79%. Abu-Freha et al.[7] described a 25% reduction in colorectal cancer diagnoses and a 33% reduction in gastric cancer diagnoses when comparing 2020 with 2019 in a regional health system. In the Brazilian setting specifically, Faria et al.[8] reported a 39% reduction in endoscopic ultrasound procedures in a high-volume endoscopy unit in São Paulo, from 475 procedures in the pre-pandemic year to 289 in the first pandemic year, with a compensatory increase in the proportion of interventional relative to diagnostic procedures. The comparatively smaller and non-significant reduction observed in Maranhão may reflect several non-mutually exclusive factors, including differences in local service reorganization, a lower baseline procedure volume that limits statistical power to detect a level change, regional heterogeneity in the timing and intensity of restrictive measures, or a true difference in the magnitude of disruption, which this ecological, aggregate-level design cannot distinguish.

Regarding gastrointestinal cancer mortality, the segmented negative binomial model showed a statistically significant reduction in the death count coinciding with the pandemic onset, with a level-change incidence rate ratio (IRR) of 0.780 ($p<0.001$), while the post-pandemic slope change was not significant ($p=0.950$). This finding should not be interpreted as evidence that the pandemic reduced gastrointestinal cancer mortality. An abrupt drop in a mortality time series coinciding with a major disruption to the health information system is more plausibly explained by delays or under-registration in the Mortality Information System (SIM) during the acute phase of the pandemic, when vital registration, death certification, and data processing capacity were themselves under strain, than by a genuine decline in underlying cancer deaths. This interpretation is consistent with the broader disruption to non-COVID health services described internationally during the same period. Ogallo et al.[9] compared interrupted time series and Prophet-based approaches for quantifying the impact of COVID-19 on essential health services and demonstrated that abrupt, model-detectable shifts in routine health service indicators during

the pandemic period frequently reflected changes in reporting and service delivery rather than proportional changes in underlying disease burden, which reinforces the need for caution when interpreting isolated level changes in administrative or registry-based mortality series.

The cross-correlation function analysis performed between the differenced endoscopy and mortality series did not reveal a coherent or biologically plausible pattern of lagged association at any of the lags tested. The number of lags exceeding the significance bounds was compatible with the number expected by chance alone given the multiple comparisons involved, and no consistent direction or delay pattern emerged. This result should not be read as evidence against a causal pathway linking reduced endoscopic screening to later cancer mortality, since such a pathway, if present, would plausibly unfold over a period longer than the analytic window available in this study (2018 to 2024, with only limited follow-up after the pandemic onset). Rather, it indicates that no such association is yet detectable with the present data and analytic approach, a conclusion in line with the broader literature suggesting that the mortality consequences of pandemic-related screening disruption are expected to accumulate and become measurable only in the medium to long term.

Several limitations inherent to the ecological design of this study must be acknowledged. First, all analyses were conducted on aggregated, population-level monthly data, so individual-level associations between screening interruption and cancer outcomes cannot be inferred from these results, and any such inference would constitute an ecological fallacy. Second, the possibility of reverse causation or bidirectional influence between reduced procedure volume and other pandemic-related health system changes cannot be excluded using this design. Third, unmeasured confounding is likely, since the pandemic period coincided with numerous concurrent changes in the health system, including reallocation of hospital beds and personnel, changes in patient care-seeking behavior, and evolving public health restrictions, none of which could be individually adjusted for in this time series framework. Fourth, temporal aggregation at the monthly level may obscure shorter-term fluctuations, particularly during the earliest weeks of the pandemic when restrictive measures were most abrupt. For these reasons, the associations described here should be interpreted as descriptive and hypothesis-generating rather than causal.

Taken together, these results suggest that, in Maranhão, the COVID-19 pandemic was not associated with a statistically robust immediate disruption in the volume of upper gastrointestinal

endoscopy or colonoscopy when the analysis is anchored to the legally defined onset of the state public health emergency, although the sensitivity analysis indicates that this conclusion is dependent on the chosen breakpoint. The observed reduction in registered gastrointestinal cancer deaths at the same time point most likely reflects disruption to vital registration rather than a true decline in cancer mortality, and no lagged association between screening volume and mortality was detectable within the study period. Continued monitoring of both endoscopic procedure volume and gastrointestinal cancer mortality in Maranhão, extending beyond the period analyzed here, will be necessary to determine whether the reduction in screening activity translates into a measurable increase in mortality or in the diagnosis of more advanced disease stages over the medium and long term.

REFERENCES

1. Bray F, Ferlay J, Soerjomataram I, Siegel RL, Torre LA, Jemal A. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin. 2018;68(6):394-424.

2. Leso HD, Moraes JB, Amorim I, Carmo JPM. Gastrointestinal tract cancer epidemiology between 1999 and 2019 in Itumbiara, Goias, Brazil. Res Soc Dev. 2022;11(15):e483111537540. https://doi.org/10.33448/rsd-v11i15.37540

3. Taveira LN, Ricci TC, Queiroz MTA, Zeitune JMR. Upper gastrointestinal endoscopy in the Brazilian public health system: quantitative analysis by states and regions of the country. GED Gastroenterol Endosc Dig. 2011;30(4):142-7.

4. Lima Kubo HK, Campiolo EL, Ochikubo GT, Batista G. Impact of the COVID-19 pandemic on health services: a literature review. InterAm J Med Health. 2020;3. https://doi.org/10.31005/iajmh.v3i0.140

5. Assunção VR, et al. The influence of the COVID-19 pandemic on patients in an oncology outpatient clinic: a qualitative study. Rev Ibero-Am Humanid Cienc Educ. 2024;10(5):3197-212.

6. Mazidimoradi A, Tiznobaik A, Salehiniya H. Impact of the COVID-19 Pandemic on Colorectal Cancer Screening: a Systematic Review. J Gastrointest Cancer. 2022;53(3):730-744. https://doi.org/10.1007/s12029-021-00679-x

7. Abu-Freha N, Hizkiya R, Abu-Abed M, Michael T, Jacob BM, Rouvinov K, et al. The impact of the COVID-19 pandemic on colorectal and gastric cancer diagnosis, disease stage and mortality. Front Med. 2022;9:954878. https://doi.org/10.3389/fmed.2022.954878

8. Faria ATR, Omairi TW, Krubniki BR, Silva BL, Micelli-Neto O, Taglieri E, Ardengh JC. The impact of the COVID-19 pandemic on endoscopic ultrasound procedures in a high-volume endoscopy unit in Brazil. Arq Gastroenterol. 2023;60(3):364-372. https://doi.org/10.1590/s0004-2803.230302023-75

9. Ogallo W, Wanyana I, Tadesse GA, Wanjiru C, Akinwande V, Kabwama S, et al. Quantifying the impact of COVID-19 on essential health services: a comparison of interrupted time series analysis using Prophet and Poisson regression models. J Am Med Inform Assoc. 2023;30(4):634-642. https://doi.org/10.1093/jamia/ocac223